

Counting ℓ -adic sheaves

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to Gérard Laumon, on the occasion of his sixtieth birthday

1 Introduction

1.1. Let X_0 be a projective, smooth, absolutely connected curve of genus g over a finite field $\mathbf{F}_q$ of characteristic p , and let X be obtained from it by extension of scalars to an algebraic closure $\mathbf{F}$ of $\mathbf{F}_q$:

$$\begin{array}{ccc} X & \xrightarrow{\quad} & X_0 \\ \downarrow & & \downarrow \\ \mathrm{Spec}(\mathbf{F}) & \xrightarrow{\quad} & \mathrm{Spec}(\mathbf{F}_q). \end{array} \quad (1.1.1)$$

The endomorphism of X_0 which is the identity on the underlying topological space and is $f \mapsto f^q$ on the structure sheaf is an endomorphism of the $\mathbf{F}_q$ -scheme X_0 . We shall denote by Frob the endomorphism of the $\mathbf{F}$ -scheme X obtained from it by extension of scalars. This is the Frobenius endomorphism of X . Fix a prime number $\ell \neq p$, an algebraic closure $\overline{\mathbf{Q}}_\ell$ of $\mathbf{Q}_\ell$, and let E be the set of isomorphism classes of irreducible smooth $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on X . Pullback by Frob induces a permutation of E . We shall denote it by V .

1.2. In the article [Dr], which remains as mysterious to me as it was 31 years ago, Drinfeld computes the number of fixed points of $V : E \rightarrow E$.

A rank-one $\overline{\mathbf{Q}}_\ell$ -sheaf L_0 on $\mathrm{Spec}(\mathbf{F}_q)$ is determined up to isomorphism by the unit λ of $\overline{\mathbf{Q}}_\ell$ such that the geometric Frobenius $\mathrm{Fr} \in \mathrm{Gal}(\mathbf{F}/\mathbf{F}_q)$ acts by multiplication by λ on the fiber of L_0 at the geometric point $\mathrm{Spec}(\mathbf{F})$. The $\mathbf{F}_q$ -twist of F_0 on X_0 by L_0 is the tensor product with the inverse image of L_0 on X_0 . By abuse of language, we shall also say “ $\mathbf{F}_q$ -twist by λ ”.

Drinfeld uses the fact that the isomorphism class of a smooth $\overline{\mathbf{Q}}_\ell$ -sheaf F on X is fixed by Frob^* if and only if F is the inverse image of a $\overline{\mathbf{Q}}_\ell$ -sheaf F_0 on X_0 , and that, if F is irreducible, F_0 is unique up to $\mathbf{F}_q$ -twist. The problem solved in [Dr] becomes that of counting smooth $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on X_0 , taken up to $\mathbf{F}_q$ -twist, and considering only those which are irreducible and remain so after inverse image on X .

In 1981, one almost had the correspondence between irreducible smooth $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on X_0 and everywhere unramified cuspidal automorphic representations for $\mathrm{GL}(2, k(X_0))$. By means of this correspondence, Drinfeld reduced the problem to an application of the trace formula for $\mathrm{GL}(2)$. This reduction shows that the number sought is independent of ℓ . The formula obtained by Drinfeld has the following miraculous properties (A) and (B).

For $n \geq 1$, let N_n be the number of fixed points of the n -th iterate $V^n : E \rightarrow E$ of V . Computing N_n is the preceding problem, with $X_0/\mathbf{F}_q$ replaced by $(X_0 \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n})/\mathbf{F}_{q^n}$.

(A) The function $n \mapsto N_n$ has the form

$$N_n = \sum_i a_i \beta_i^n, \quad (1.2.1)$$

for suitable integers a_i and suitable Weil numbers β_i .

The β_i are monomials in q and in the eigenvalues of the endomorphism Frob^* of $H^1(X, \overline{\mathbf{Q}}_\ell)$. With the genus g fixed, which monomials β_i appear, and with which multiplicities a_i , does not depend on the curve considered.

The trace formula expresses N_n as the sum of several terms. Taken separately, these terms do not all have the form (1.2.1).

Let Σ be a compact Riemann surface of genus g . Consider local systems of complex vector spaces on Σ . Let M be the moduli space of those which are irreducible of rank 2. The space M and the set E are empty if $g = 0$ or 1 . Otherwise, M is a connected complex symplectic variety. Its complex dimension is therefore an even integer $2N$.

(B) In (1.2.1), the dominant term is q^N : one of the β_i is q^N , its multiplicity a_i is 1, and the other β_i satisfy $|\beta_i| < q^N$.

1.3. Formula (1.2.1) is reminiscent of a Lefschetz trace formula in which, in good cases, the number of fixed points of the iterates T^n ($n \geq 1$) of an endomorphism T of a space S is

$$\text{Tr}(T^{*n}, H_\gamma^*(S)) := \sum_i (-1)^i \text{Tr}(T^{*n}, H_\gamma^i(S)). \quad (1.3.1)$$

The “?” is there to recall that one must consider a cohomology theory with support conditions. The support conditions to impose depend on the behavior of T at infinity. For example, if for an exhaustion function f on S one has $f(T(x)) > f(x)$ (respectively $f(T(x)) < f(x)$) for $f(x)$ sufficiently large, then the cohomology to consider is compactly supported cohomology (respectively ordinary cohomology).

Assume that H_γ^* is finite-dimensional, so that (1.3.1) makes sense. The right-hand side of (1.3.1) is then of the form (1.2.1): it is $\sum a(\beta)\beta^n$, where β runs through the eigenvalues of T^* and the integer $a(\beta)$ is the alternating sum of the multiplicities of β as an eigenvalue of the endomorphisms T^* of the $H_\gamma^i(S)$.

Unfortunately, I have no idea how to regard the set E of isomorphism classes of irreducible smooth $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on X as a “space” having a cohomology $H_\gamma^*(E)$ such that one could hope for the number of fixed points of V^n to be given by a formula of type (1.3.1).

1.4. Even if we do not know how to think geometrically about E , if $f \in E$ is the isomorphism class of F , we know what the formal completion $\widehat{E}_f$ of E at f should be. A finite-dimensional local $\overline{\mathbf{Q}}_\ell$ -algebra Λ is augmented to $\overline{\mathbf{Q}}_\ell$. A deformation of F over $\text{Spec}(\Lambda)$ is a smooth $\overline{\mathbf{Q}}_\ell$ -sheaf F_Λ on X , equipped with a Λ -module structure and an isomorphism $F_\Lambda \otimes_\Lambda \overline{\mathbf{Q}}_\ell \xrightarrow{\sim} F$, such that at a point x (and hence at every point x of X), $(F_\Lambda)_x$ is a free Λ -module. The functor, in $\text{Spec}(\Lambda)$, of isomorphism classes of deformations of F over $\text{Spec}(\Lambda)$ is pro-representable. Let R_F denote the complete local $\overline{\mathbf{Q}}_\ell$ -algebra with residue field $\overline{\mathbf{Q}}_\ell$ such that $\text{Spf}(R_F)$ pro-represents it. There are no obstructions to deformations, and R_F is therefore isomorphic to a formal power-series algebra $\overline{\mathbf{Q}}_\ell[[t_1, \dots, t_m]]$. Because F is irreducible, its only automorphisms are multiplications by $\lambda \in \overline{\mathbf{Q}}_\ell^*$. They extend to every deformation. If F and F' are two representatives of the isomorphism class f , two isomorphisms from F to F' therefore induce the same isomorphism between R_F and $R_{F'}$: $\widehat{E}_f := \text{Spf}(R_F)$ depends only on f , up to a unique isomorphism.

The Zariski tangent space of $\widehat{E}_f$ at f is $H^1(X, \text{End}(F))$. The symmetric bilinear form $\text{Tr}(fg)$ on $\text{End}(F)$ is a self-duality. It induces on $H^1(X, \text{End}(F))$ a symplectic form with values in $H^2(X, \overline{\mathbf{Q}}_\ell) = \overline{\mathbf{Q}}_\ell(-1)$. The same construction gives the symplectic structure of M over $\mathbb{C}$. The formal completion $\widehat{E}_f$ is therefore smooth of even dimension, the same as that of M .

The pullback functor by $\text{Frob} : X \rightarrow X$ transforms deformations of F into deformations of

$\text{Frob}^* F$. It therefore induces a morphism

$$V_f^\wedge : \widehat{E}_f \longrightarrow \widehat{E}_{V(f)}.$$

The argument which shows that, because Frob induces an equivalence of étale sites, V is bijective also shows that the $V_f^\wedge$ are isomorphisms. Another proof is the following. Let ω be the symplectic form defined above on the tangent space at f . If $[t] \in H^1(X, \text{End}(F))$ is the class of the tangent vector t at F , the class of $dV_f^\wedge(t)$ in $H^1(X, \text{End}(\text{Frob}^* F))$ is $\text{Frob}^*([t])$. For two tangent vectors t_1 and t_2 , $\omega(dV_f^\wedge(t_1), dV_f^\wedge(t_2))$ is therefore the inverse image, by Frob^* , of $\omega(t_1, t_2) \in \overline{\mathbf{Q}}_\ell(-1)$, identified with $H^2(X, \overline{\mathbf{Q}}_\ell)$. Because $\text{Frob} : X \rightarrow X$ has degree q , the endomorphism Frob^* of $H^2(X, \overline{\mathbf{Q}}_\ell)$ is multiplication by q , and

$$\omega(dV_f^\wedge(t_1), dV_f^\wedge(t_2)) = q\omega(t_1, t_2),$$

which may also be written $(dV_f^\wedge)^*\omega = q\omega$. Thus the differential $dV_f^\wedge$ is an isomorphism.

The map $dV_f^\wedge$ between Zariski tangent spaces is identified with

$$\text{Frob}^* : H^1(X, \text{End}(F)) \longrightarrow H^1(X, \text{End}(\text{Frob}^* F)).$$

If f is a fixed point of V , then F is the inverse image of F_0 on X_0 . We know that F_0 can be chosen pure, and then $\text{End}(F_0)$ is pure of weight 0. The eigenvalues of $dV_f^\wedge$ on the Zariski tangent space of $\widehat{E}_f$ at f are therefore Weil numbers of weight 1. In particular, no eigenvalue of $dV_f^\wedge$ at f is equal to 1. This is what makes it reasonable that a Lefschetz-type formula (1.3.1) should be able to count the fixed points of V^n , all taken with multiplicity one.

1.5. If Y_0 is a scheme of finite type over $\mathbf{F}_q$, the Frobenius endomorphism Frob of $Y := Y_0 \otimes_{\mathbf{F}_q} \mathbf{F}$ induces a permutation Frob of $Y(\mathbf{F})$. The preceding discussion shows that the nature of (E, V) is different from that of (Y, Frob) :

- (i) E is “over $\overline{\mathbf{Q}}_\ell$ ”, hence an object of characteristic 0.
- (ii) The $V_f^\wedge$ are isomorphisms. One would like to say that V has “degree one”.
- (iii) The fixed points of V are isolated because the eigenvalues of dV at a fixed point are Weil numbers of weight 1. Those of Frob are isolated because $d\text{Frob} = 0$.
- (iv) If E has dimension $2N$, in the sense that the $\widehat{E}_f$ do, then the order of magnitude of the number of fixed points of V^n is $(q^n)^N$, rather than $(q^n)^{2N}$.

1.6. The moduli space M is a complex algebraic variety. Contrary to what a hasty analogy might suggest, E is not naturally the space of $\overline{\mathbf{Q}}_\ell$ -points of an algebraic variety over $\overline{\mathbf{Q}}_\ell$. Let $x \in X(\mathbf{F})$. The set E is identified with the set of isomorphism classes of continuous irreducible representations V of $\pi_1(X, x)$, of dimension 2 over $\overline{\mathbf{Q}}_\ell$. As coordinates on E , one would like to take the $\text{Tr}(\gamma, V)$ for $\gamma \in \pi_1(X, x)$. In the complex case, this construction gives M its “Betti” structure as a complex algebraic variety. Here, since $\pi_1(X, x)$ is a profinite group, every representation V admits an invariant lattice, and the $\text{Tr}(\gamma, V)$ take values in the valuation ring $\overline{\mathbf{Z}}_\ell$ of the valued field $\overline{\mathbf{Q}}_\ell$.

The viewpoint that E is a “space of representations of $\pi_1(X, x)$ ” does not help either in understanding why the number of fixed points of V is independent of ℓ . Another mystery is the following: why, with the genus g fixed, does the number of fixed points of V admit a uniform description in terms of $H^1(X, \overline{\mathbf{Q}}_\ell)$, its symplectic structure with values in $\overline{\mathbf{Q}}_\ell(-1)$, and its endomorphism Frob^* ,

whereas $\pi_1(X, x)$ is a quotient of the profinite completion of the topological π_1 of Σ , with that quotient depending on X ?

1.7. Let E_r be the set of isomorphism classes of irreducible smooth $\overline{\mathbf{Q}}_\ell$ -sheaves of rank r on X . Thanks to Lafforgue [L], the number of fixed points of the bijection $\text{Frob}^* : E_r \rightarrow E_r$ has an automorphic interpretation. For $S_0 \subset X_0$ a finite set of closed points of X_0 , and $S = S_0 \otimes_{\mathbf{F}_q} \mathbf{F}$, one may also consider irreducible smooth $\overline{\mathbf{Q}}_\ell$ -sheaves on $X - S$, with prescribed ramification at each $s \in S$. In all cases that have been computed, the miracles (A) (B) discovered by Drinfeld persist, *mutatis mutandis*. The purpose of this talk is to give a panorama of what is known.

2 What should be counted?

2.1. Fix an integer $r \geq 1$, and let $X_0, S_0/\mathbf{F}_q$ and $X, S/\mathbf{F}$ be as in 1.1 and 1.7. For $s \in S$, let $X_{(s)}$ be the henselization of X at s , and let $X_{(s)}^* := X_{(s)} - \{s\}$. Suppose given, for each $s \in S$, a smooth $\overline{\mathbf{Q}}_\ell$ -sheaf $R(s)$ of rank r on $X_{(s)}^*$. Only its isomorphism class matters. The morphism $\text{Frob} : X \rightarrow X$ induces a morphism, still denoted Frob , from $X_{(s)}^*$ to $X_{(\text{Frob}(s))}^*$. We assume that, for every s in S , $R(s)$ is isomorphic to the inverse image of $R(\text{Frob}(s))$ by Frob :

$$R(s) \simeq \text{Frob}^*(R(\text{Frob}(s))). \quad (2.1.1)$$

Let $R := (R(s))_{s \in S}$ be the family of the $R(s)$, and denote by $E(R)$ the set of isomorphism classes of irreducible smooth $\overline{\mathbf{Q}}_\ell$ -sheaves F of rank r on $X - S$, such that for every $s \in S$, the inverse image $F|_{X_{(s)}^*}$ of F on $X_{(s)}^*$ is isomorphic to $R(s)$. Hypothesis (2.1.1) ensures that the pullback functor by Frob induces a permutation of $E(R)$. We shall denote this permutation by V .

Our choice of “ V ”, the first letter of “Verschiebung”, is due to an analogy with the case of certain abelian varieties, explained in 2.2 below. In this analogy, q is classically a prime number rather than a power of a prime number.

2.2. For every scheme Y over $\mathbf{F}_q$, denote by Frob_Y the endomorphism of the $\mathbf{F}_q$ -scheme Y which is the identity on the underlying topological space and is given on the structure sheaf by $\text{Frob}_Y^*(f) = f^q$. This is the Frobenius endomorphism of Y .

Take for Y the abelian variety $\text{Pic}^0(X_0)$ over $\mathbf{F}_q$. We get

$$\text{Frob}_{\text{Pic}^0(X_0)} : \text{Pic}^0(X_0) \longrightarrow \text{Pic}^0(X_0). \quad (2.2.1)$$

The Jacobian $\text{Pic}^0(X_0)$ represents the following functor on $\mathbf{F}_q$ -schemes: the sheaf (for the Zariski topology, or for the étale topology, which comes to the same thing) associated with the presheaf which attaches to S the group $\text{Pic}^0(S \times_{\mathbf{F}_q} X_0)$ of invertible sheaves on $S \times_{\mathbf{F}_q} X_0$, of degree zero on each fiber of the projection to S . If $f : Y_0 \rightarrow X_0$ is a morphism, inverse image of invertible sheaves by f defines a morphism $f^* : \text{Pic}^0(X_0) \rightarrow \text{Pic}^0(Y_0)$. Take for f the endomorphism Frob_{X_0} of X_0 . The endomorphism $\text{Frob}_{X_0}^*$ of $\text{Pic}^0(X_0)$ deserves to be called V , since it satisfies

$$\text{Frob}_{X_0}^* \circ \text{Frob}_{\text{Pic}^0(X_0)} = \text{Frob}_{\text{Pic}^0(X_0)} \circ \text{Frob}_{X_0}^* = q. \quad (2.2.2)$$

Proof of (2.2.2). Interpret (2.2.2) as two equalities between endomorphisms of the functor represented by $\text{Pic}^0(X_0)$. For every contravariant functor G on the category of $\mathbf{F}_q$ -schemes, the Frobenius endomorphism Frob_G of G is defined by

$$\text{Frob}_G(S) : G(S) \rightarrow G(S), \quad \text{namely } G(\text{Frob}_S).$$

If G represents Y , Frob_G is induced by Frob_Y .

Take for G the functor $S \mapsto \text{Pic}(S \times_{\mathbf{F}_q} X_0)$ of isomorphism classes of invertible sheaves on $S \times_{\mathbf{F}_q} X_0$. The endomorphism $\text{Frob}_G(S)$ of $\text{Pic}(S \times_{\mathbf{F}_q} X_0)$ is inverse image of isomorphism classes of invertible sheaves by $\text{Frob}_S \times \text{Id}_{X_0} : S \times_{\mathbf{F}_q} X_0 \rightarrow S \times_{\mathbf{F}_q} X_0$. If one composes this inverse image with inverse image by Frob_{X_0} , one obtains inverse image by $\text{Frob}_{S \times_{\mathbf{F}_q} X_0}$, which is nothing other than raising to the q -th tensor power. Considering only invertible sheaves of degree zero on the fibers of the projection to S , and passing from the resulting presheaf to the associated sheaf, we obtain (2.2.2).

If now one extends scalars from $\mathbf{F}_q$ to $\mathbf{F}$, then Frob_{X_0} becomes the endomorphism Frob of X , $\text{Pic}^0(X_0)$ becomes $\text{Pic}^0(X)$, and the endomorphism $\text{Frob}_{X_0}^*$ of $\text{Pic}^0(X_0)$ becomes the endomorphism of $\text{Pic}^0(X)$ given by “inverse image by Frob ”.

2.3. The questions we ask are the following.

- (i) Compute, for $n \geq 1$, the number $N_n(R)$ of fixed points of the n -th iterate V^n of the permutation V of $E(R)$. Notice that replacing (X_0, S_0) by $(X_0, S_0) \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}$ replaces V by V^n .
- (ii) Determine whether the miracles 1.2 (A) (B) persist.
- (iii) If so, why?

Let F be a smooth $\overline{\mathbf{Q}}_\ell$ -sheaf on $X - S$. As in 1.2, we have:

Lemma 2.4. (i) *For the isomorphism class of F to be fixed under Frob^* , it is necessary and sufficient that F be isomorphic to the inverse image on X of a $\overline{\mathbf{Q}}_\ell$ -sheaf F_0 on X_0 .*

Furthermore, if F is irreducible,

- (ii) *such an F_0 is unique up to $\mathbf{F}_q$ -twist, and*
- (iii) *the $\mathbf{F}_q$ -twists of F_0 by $\lambda \in \overline{\mathbf{Z}}_\ell^* (1.2)$ are pairwise non-isomorphic.*

By 2.4, the number of fixed points of the permutation V of $E(R)$ is the number of classes, modulo $\mathbf{F}_q$ -twist, of smooth $\overline{\mathbf{Q}}_\ell$ -sheaves F_0 of rank r on $X_0 - S_0$, such that

- (a) the inverse image F of F_0 on $X - S$ is irreducible;
- (b) for each $s \in S$, $F|_{X_{(s)}^*}$ is isomorphic to $R(s)$.

Let $D_0 = \sum n_i x_i$ be a divisor of degree one on $X_0 - S_0$. Regard F_0 as a Galois representation (see 2.7), let F_{x_i} be a Frobenius at x_i , and put

$$\langle D_0, F_0 \rangle := \prod_i \det(F_{x_i}, F_0)^{n_i}.$$

If F'_0 is the $\mathbf{F}_q$ -twist of F_0 by λ , then

$$\langle D_0, F'_0 \rangle = \lambda^r \langle D_0, F_0 \rangle.$$

By 2.4 (ii) and (iii), it therefore comes to the same thing to count the F_0 satisfying (a) and (b) up to $\mathbf{F}_q$ -twist, or to count, up to isomorphism, only those that in addition satisfy $\langle D_0, F \rangle = 1$, and then divide by r . For an automorphic counterpart, see the remark following 2.6.

2.5. Put $K := k(X_0)$. For each closed point x of X_0 , let $k(x)$ be the residue field of x , let $\deg(x) := [k(x) : \mathbf{F}_q]$ be its degree, let K_x be the completion of K at x , and let O_x be the ring of

the valuation v_x of K_x . Let A be the ring of adèles of K . The sum $v := \sum \deg(x)v_x : A^* \rightarrow \mathbb{Z}$ is trivial on K^* . We have $\|a\| = q^{-v(a)}$.

By Lafforgue, F_0 corresponds to a cuspidal automorphic representation π of $\mathrm{GL}(r, A)$. If F'_0 is the $\mathbf{F}_q$ -twist of F_0 by λ , the corresponding representation π' is the $\mathbf{F}_q$ -twist of π by λ , defined as follows. As a representation, it is the tensor product of π by the one-dimensional representation $g \mapsto \lambda^{v \det g}$. As a space of functions on $\mathrm{GL}(r, A)$, it is the space of products of f in π by the character $g \mapsto \lambda^{v \det g}$.

Translated into automorphic language, problem 2.3 (i) is that of counting, up to $\mathbf{F}_q$ -twist, the cuspidal automorphic representations π of $\mathrm{GL}(r, A)$, unramified outside S_0 , and such that

- (a') For all $n \geq 1$, π remains cuspidal after base change from K to $K \otimes \mathbf{F}_{q^n}$. This condition expresses the irreducibility 2.4 (a) of F . It is enough to check it for n dividing r .
- (b') For each $s_0 \in S_0$ and $s \in S$ above s_0 , the local component π_{s_0} of π satisfies a condition dictated by $R(s)$ and by the local Langlands correspondence.

Lemma 2.6. *If the cuspidal automorphic representation π satisfies (a'), then the $\mathbf{F}_q$ -twists of π are pairwise distinct.*

This is the translation of 2.4 (iii).

If d is an idèle satisfying $v(d) = 1$, it therefore comes to the same thing to count, up to $\mathbf{F}_q$ -twist, the π satisfying (a') (b'), or to count only those whose central character ω_π satisfies $\omega_\pi(d) = 1$, and then divide by r .

In finite characteristic, the notion of an automorphic representation is purely algebraic. This makes it possible to take the automorphic representations π as having coefficients in $\overline{\mathbf{Q}}_\ell$, rather than in $\mathbb{C}$. If this is done, π_{s_0} is an irreducible admissible $\overline{\mathbf{Q}}_\ell$ -representation of $\mathrm{GL}(r, K_{s_0})$, and the local Langlands correspondence associates to it an isomorphism class of continuous F -semisimple $\overline{\mathbf{Q}}_\ell$ -representations of dimension r of the local Weil group $W(\overline{K}_{s_0}/K_{s_0})$. Condition (b') is that its restriction to the inertia subgroup be given by $\mathcal{R}(s)$ (cf. 2.4 below).

Almost all known results on question 2.3 (i) are obtained through the dictionary 2.5.

2.7. In 2.1 we used the language of $\overline{\mathbf{Q}}_\ell$ -sheaves. This spared us from having to choose base points. An equivalent language is that of Galois representations. Here is the translation.

Let $\overline{k(X)}$ be an algebraic closure of the rational function field $k(X)$ of X , and let $k(X)_{nr}$ be the largest subextension unramified outside S . The Galois group $\mathrm{Gal}(k(X)_{nr}/k(X))$ is the fundamental group of X at the geometric point $\bar{\eta} := \mathrm{Spec}(\overline{k(X)})$ of X : the functor “fiber at $\bar{\eta}$ ” is an equivalence of categories from smooth $\overline{\mathbf{Q}}_\ell$ -sheaves (respectively irreducible smooth $\overline{\mathbf{Q}}_\ell$ -sheaves of rank r) on $X - S$ to linear $\overline{\mathbf{Q}}_\ell$ -representations of $\mathrm{Gal}(k(X)_{nr}/k(X))$ (respectively irreducible representations of rank r).

The scheme $X_{(s)}^*$ is the spectrum of an extension $k(X_{(s)}^*)$ of $k(X)$. If one chooses an embedding of this extension in $\overline{k(X)}$, the corresponding inertia group $I_s := \mathrm{Gal}(\overline{k(X)}/k(X_{(s)}^*))$ is likewise the fundamental group of $X_{(s)}^*$ at $\bar{\eta}$. It maps to $\pi_1(X, \bar{\eta})$:

$$\pi_1(X_{(s)}^*, \bar{\eta}) = I_s = \mathrm{Gal}(\overline{k(X)}/k(X_{(s)}^*)) \longrightarrow \mathrm{Gal}(k(X)_{nr}/k(X)) = \pi_1(X, \bar{\eta}). \quad (2.7.1)$$

The fiber at $\bar{\eta}$ of $\mathcal{R}(s)$ is a representation ρ_s of I_s .

Let $\mathcal{F}$ be a smooth $\overline{\mathbf{Q}}_\ell$ -sheaf on $X - S$, with inverse image $\mathcal{F}|_{X_{(s)}^*}$ on $X_{(s)}^*$. The fibers at the geometric point $\bar{\eta}$ are respectively a representation of $\pi_1(X - S, \bar{\eta}) = \mathrm{Gal}(k(X)_{nr}/k(X))$, and its restriction to $\pi_1(X_{(s)}^*, \bar{\eta}) = I_s$ by (2.4.1). The condition that $\mathcal{F}|_{X_{(s)}^*}$ be isomorphic to $\mathcal{R}(s)$ is equivalent to requiring this restriction to I_s to be isomorphic to ρ_s .

Condition (2.1.1) is equivalent to saying that the ρ_s , for s lying above s_0 , come from a representation of a decomposition group $D_{s_0} \subset \text{Gal}(k(\bar{X})/k(X_0))$ at s_0 . It implies that the representations ρ_s are quasi-unipotent.

2.8. I hope that problem 2.3 (i) is more manageable when a smooth $\bar{\mathbf{Q}}_\ell$ -sheaf $\mathcal{F}$ such that $\mathcal{F}|_{X_{(s)}^*} \sim \mathcal{R}(s)$ for every $s \in S$ is automatically irreducible. This is the case if the following condition is satisfied. Consider an integer $0 < r' < r$, and, for each $s \in S$, a smooth sub- $\bar{\mathbf{Q}}_\ell$ -sheaf $\mathcal{R}'(s)$ of rank r' of $\mathcal{R}(s)$. Put $\mathcal{R}''(s) := \mathcal{R}(s)/\mathcal{R}'(s)$. The condition is the following.

(2.8.1) For all r' and all $\mathcal{R}'(s)$, either there is no smooth $\bar{\mathbf{Q}}_\ell$ -sheaf $\mathcal{F}'$ of rank r' on $X - S$ such that $\mathcal{F}'|_{X_{(s)}^*} \sim \mathcal{R}'(s)$ for every $s \in S$, or there is no smooth $\bar{\mathbf{Q}}_\ell$ -sheaf $\mathcal{F}''$ of rank $r - r'$ on $X - S$ such that $\mathcal{F}''|_{X_{(s)}^*} \sim \mathcal{R}''(s)$ for every $s \in S$.

Condition (2.8.1) is satisfied if one of the $\mathcal{R}(s)$ is irreducible. For other cases, see 2.10.4, 2.11, and 2.12.

2.9. For every field k of characteristic p or 0, containing all roots of unity of order prime to p , let us set

$$\widehat{\mathbf{Z}}^{p'}(1)(k) := \varprojlim \mu_n(k). \quad (2.9.1)$$

The projective limit is taken over the set of subgroups $\mu_n(k)$ of k^* , with n prime to p , ordered by inclusion. The transition morphism from $\mu_m(k)$ to $\mu_n(k)$ is $x \mapsto x^{m/n}$. If this creates no ambiguity, we omit “ p ” and the mention of k .

For $k = \mathbf{F}$, and $\mathbf{F}_{q^k}$ the subfield of $\mathbf{F}$ with q^k elements, the subgroups $\mathbf{F}_{q^k}^*$ of $\mathbf{F}^*$ are cofinal among the $\mu_n(\mathbf{F})$. Since $(q^{kd} - 1)/(q^k - 1) = 1 + q^k + \dots + q^{k(d-1)}$, the transition morphism from $\mu_{q^{kd}-1} = \mathbf{F}_{q^{kd}}^*$ to $\mu_{q^k-1} = \mathbf{F}_{q^k}^*$ is the norm morphism, and

$$\widehat{\mathbf{Z}}(1) \xrightarrow{\sim} \varprojlim \mathbf{F}_{q^k}^*. \quad (2.9.2)$$

(the projective limit is taken with respect to the norm morphisms between the $\mathbf{F}_{q^k}^*$). For k a product $\prod k_\alpha$ of finite extensions of $\mathbf{F}_q$ and $\iota : k \rightarrow k_\alpha \xrightarrow{\sim} \mathbf{F}_{q^{k_\alpha}} \hookrightarrow \mathbf{F}$, denote by pr_ι the composite

$$\text{pr}_\iota : \widehat{\mathbf{Z}}(1) \longrightarrow \mathbf{F}_{q^k}^* \xleftarrow{\sim} k_\alpha^* \hookrightarrow \bigoplus k_\alpha^* = k^*. \quad (2.9.3)$$

For $k = \mathbf{F}_{q^k}$ and ι the identity inclusion, $\text{Gal}(\mathbf{F}/\mathbf{F}_{q^k})$ acts on $\widehat{\mathbf{Z}}(1)$, and (2.6.3) induces an isomorphism of coinvariants with $\mathbf{F}_{q^k}^*$.

The local fundamental group $\pi_1(X_{(s)}^*, \bar{\eta}) = I_s$ is an extension of $\widehat{\mathbf{Z}}(1)$ by a pro- p -group P_s . The quotient $I_s/P_s \simeq \widehat{\mathbf{Z}}(1)$ is the tame fundamental group of $X_{(s)}^*$. The morphism $\text{Frob} : X_{(s)}^* \rightarrow X_{(\text{Frob}(s))}^*$ induces a morphism between tame fundamental groups. Under the identification of both with $\widehat{\mathbf{Z}}(1)$, this morphism is multiplication by q .

Let $\mathbb{Z}_\ell(1)$ be the quotient $\varprojlim \mu_{\ell^n}(\mathbf{F})$ of $\widehat{\mathbf{Z}}(1)$. Every quasi-unipotent $\bar{\mathbf{Q}}_\ell$ -representation ρ of $\widehat{\mathbf{Z}}(1)$ is isomorphic to a sum over finite-order characters $\chi : \widehat{\mathbf{Z}}(1) \rightarrow \bar{\mathbf{Q}}_\ell^*$

$$\rho \sim \bigoplus \chi \otimes \nu_\rho(\chi), \quad (2.9.4)$$

where $\nu_\rho(\chi)$ is a unipotent representation of the quotient $\mathbb{Z}_\ell(1)$ of $\widehat{\mathbf{Z}}(1)$. Let $\lambda_\rho(\chi)$ be the following partition of the integer $\dim \nu_\rho(\chi)$: the decreasing sequence of the sizes of the Jordan blocks of the image of a generator of $\mathbb{Z}_\ell(1)$, in the representation $\nu_\rho(\chi)$. The isomorphism class of ρ is determined by the function $\chi \mapsto \lambda_\rho(\chi)$.

Suppose that the $\mathcal{R}(s)$ are tame, that is, that the corresponding representations ρ_s of the inertia groups I_s (2.7) factor through a representation of $\widehat{\mathbf{Z}}(1)$. We shall write $\chi \mapsto \lambda_s(\chi)$ for the function, constructed above, attached to this representation. With this notation, condition (2.1.1) is equivalent to

$$\lambda_{\text{Frob}(s)}(\chi) = \lambda_s(\chi^q). \quad (2.9.5)$$

2.10. If $\mathcal{L}$ is a $\overline{\mathbf{Q}}_\ell$ -sheaf of rank one on $X - S$, the restriction of $\mathcal{L}$ to $X_{(s)}^*$ defines a character l_s of the fundamental group I_s of $X_{(s)}^*$. Since $\ell \neq p$, the restriction of l_s to P_s has finite order: there exists a power P of p such that the l_s^P factor through characters, still denoted l_s^P , of $\widehat{\mathbf{Z}}(1) = I_s/P_s$. We have

$$\prod l_s^P = 1. \quad (2.10.1)$$

The analogue over $\mathbb{C}$ of (2.10.1) is that, for Σ as in 1.2 and S a finite subset of Σ , the sum over S of a small positive circle around each $s \in S$ is a boundary in $\Sigma - S$.

In rank r , (2.10.1) applied to $\bigwedge^r \mathcal{F}$ gives a necessary condition for there to exist $\mathcal{F}$ on $X - S$ satisfying $\mathcal{F}|_{X_{(s)}^*} \sim \mathcal{R}(s)$, and gives interesting cases where the hypothesis of the irreducibility criterion (2.8.1) is satisfied. Let ρ_s be the representation of I_s corresponding to $\mathcal{R}(s)$. Suppose, as in 2.9, that the representations ρ_s are tame. Let $R(s)$ be the multiset of characters of $\widehat{\mathbf{Z}}(1)$ whose sum is the semisimplification of ρ_s . With the notation of 2.9, the multiplicity in $R(s)$ of a character χ is the dimension $|\lambda_s(\chi)|$ of $v_{\rho_s}(\chi)$.

Suppose that

$$\prod_{s \in S} \prod_{\varepsilon \text{ in } R(s)} \varepsilon = 1. \quad (2.10.2)$$

By (2.10.1), this is a necessary condition for there to exist an $\mathcal{F}$ on $X - S$ such that $\mathcal{F}|_{X_{(s)}^*} \sim R_s$ for every $s \in S$. Consider the condition

(2.10.3) For all $0 < r' < r$ and all families $(R'(s))_{s \in S}$ of sub-multisets with r' elements of the multisets $R(s)$,

$$\prod_{s \in S} \prod_{\varepsilon \text{ in } R'(s)} \varepsilon \neq 1.$$

Lemma 2.11. *Condition (2.10.3) implies (2.8.1), and hence the irreducibility of every smooth sheaf $\mathcal{F}$ of rank r on $X - S$ satisfying $\mathcal{F}|_{X_{(s)}^*} \sim R_s$ for every $s \in S$.*

Remark 2.12. There exist $R(s)$ as above, giving rise to multisets $R(s)$ satisfying (2.10.2), for which (2.10.3) is false, but where condition (2.8.1) is nevertheless satisfied. Here is an example. Take $X = \mathbb{P}^1$, $r = 4$, $|S| = 3$, and semi-simple $\mathcal{R}(s)$, sums of multisets of characters $R(s)$ of the following type:

$$\{a_1, a_1, a_3, a_4\}, \{b_1, b_1, b_3, b_4\}, \{c_1, c_2, c_3, c_4\}.$$

Assume that the product of all these characters is 1 (2.10.2), and that $a_4 b_4 c_4 = 1$, so that (2.10.3) is false. If the a_i, b_i, c_i are “general”, condition (2.8.1) is satisfied, since there is no $\mathcal{F}$ of rank 3 with local monodromy given by

$$\{a_1, a_1, a_3\}, \{b_1, b_1, b_3\}, \{c_1, c_2, c_3\}.$$

The verification is by reduction to the complex case, by lifting (X, S) to characteristic 0. This reduction leads to the following lemma.

Lemma 2.13. *Let $u_1, u_3, v_1, v_3, w_1, w_2, w_3$ be in $\overline{\mathbf{Q}}_\ell^*$. Suppose that $u_1 v_1 w_i \neq 1$ for $i = 1, 2, 3$. Under this hypothesis, there are no elements U, V, W of $\text{GL}(3, \overline{\mathbf{Q}}_\ell)$, respectively conjugate to the diagonal matrices (u_1, u_1, u_3) , (v_1, v_1, v_3) , (w_1, w_2, w_3) , and such that $UVW = 1$.*

Proof. Let L be the intersection of the kernels of $U - u_1$ and $V - v_1$. Since these kernels have dimension ≥ 2 , L is nonzero. On L , U is multiplication by u_1 , V is multiplication by v_1 , and since $UVW = 1$, W is multiplication by $1/u_1v_1$. Since $1/u_1v_1$ is not an eigenvalue of W , this is impossible.

2.14. Let Σ be a compact Riemann surface of genus g . Suppose an injection of S into Σ has been chosen; through it we identify S with a finite set of points of Σ . If $\Sigma_{(s)}$ is a small disk around s , the fundamental group of $\Sigma_{(s)}^* := \Sigma_{(s)} - \{0\}$ is $\mathbb{Z}$, generated by a positive loop around s . More intrinsically, this local fundamental group is $\mathbb{Z}(1) := 2\pi i\mathbb{Z}$, with $2\pi i$ corresponding to the positive loop around s . For each $n \geq 1$, we identify $\mathbb{Z}(1)/n$ with the group of n -th roots of unity in $\mathbb{C}$ by $z \mapsto \exp(z/n)$. Passing to the limit gives an isomorphism of $\widehat{\mathbb{Z}}^{p'}(1)(\mathbb{C})$ with the projective limit of the $\mathbb{Z}(1)/n$ for n prime to p .

A complex representation ρ of $\mathbb{Z}(1)$ is isomorphic to a sum over characters $\chi : \mathbb{Z}(1) \rightarrow \mathbb{C}^*$

$$\rho \sim \bigoplus \chi \otimes \nu(\chi), \quad (2.14.1)$$

where $\nu(\chi)$ is a unipotent representation of $\mathbb{Z}(1)$. Define the partition $\lambda(\chi)$ of the integer $\dim \nu(\chi)$ as in 2.9. The isomorphism class of ρ is determined by the function $\chi \mapsto \nu(\chi)$. We shall only have to consider the case where the χ appearing in (2.14.1) have finite order prime to p , and therefore may be viewed as complex characters of $\widehat{\mathbb{Z}}(1)^{p'}(\mathbb{C})$.

If T is a family $(T_s)_{s \in S}$ of isomorphism classes of complex representations of $\mathbb{Z}(1)$, we denote by $M_{\mathbb{C}}(T)$ the moduli space of irreducible complex local systems of rank r on $\Sigma - S$, with local monodromy T_s at s . This moduli space has two natural structures of algebraic variety, called Betti and de Rham; only the underlying topological space will matter to us.

Choose

(2.14.2) an isomorphism between the groups of roots of unity of order prime to p in $\overline{\mathbb{Q}}_{\ell}$ and in $\mathbb{C}$.

(2.14.3) an isomorphism between the groups of roots of unity of order prime to p in $\mathbb{F}$ and in $\mathbb{C}$.

These data are exactly what is needed to attach to an isomorphism class of quasi-unipotent ℓ -adic representations of $\widehat{\mathbb{Z}}(1)$ an isomorphism class of complex representations of $\mathbb{Z}(1)$, as one checks by comparing 2.9 and 2.14.

If the $\mathcal{R}(s)$ are tame, giving rise to representations ρ_s of $\widehat{\mathbb{Z}}(1)$, we write $\mathcal{R}^*$ for the corresponding family of complex representations $\mathcal{R}^*(s)$ of $\mathbb{Z}(1)$, and we say that the moduli space $M_{\mathbb{C}}(\mathcal{R}^*)$ corresponds to $E(\mathcal{R})$, or is the complex analogue of $E(\mathcal{R})$.

Conjecture 2.15. *Under the hypotheses and with the notation of 2.14,*

(i) *as a function of $n \geq 1$, the number $N_n(\mathcal{R})$ of fixed points of V^n acting on $E(\mathcal{R})$ is of the form*

$$N_n(\mathcal{R}) = \sum a_i \beta_i^n \quad (2.11.1)$$

for suitable integers a_i and suitable numbers β_i ;

(ii) *the sum of the a_i is equal to the Euler-Poincaré characteristic of the moduli space $M_{\mathbb{C}}(\mathcal{R}^*)$ corresponding to $E(\mathcal{R})$;*

(iii) *as in 1.2 (B), the dominant term in (2.15.1) is q to the power $\dim_{\mathbb{C}}(M_{\mathbb{C}}(\mathcal{R}^*))/2$.*

As will be explained in 6.6, if $g > 0$, then $\chi(M_{\mathbb{C}}(\mathcal{R}^*)) = 0$. For a refinement of 2.15 (i) (ii), see 6.3 and 6.7.

Using the analogy between the exponential and Artin-Schreier sheaves, it is possible to define complex analogues for certain wild representations ρ_s . Rather than moduli spaces of local systems on $\Sigma - S$, the complex analogues are moduli spaces of vector bundles with algebraic connection on $\Sigma - S$, with prescribed formal completions at each $s \in S$. I conjecture that 2.11 remains valid in this more general setting.

Variant 2.16. Let ρ and ρ' be two quasi-unipotent ℓ -adic representations $\rho, \rho' : \widehat{\mathbb{Z}}(1) \rightarrow \mathrm{GL}(r, \overline{\mathbb{Q}}_\ell)$ (respectively two complex representations $\rho, \rho' : \mathbb{Z}(1) \rightarrow \mathrm{GL}(r, \mathbb{C})$). We shall say that the isomorphism class of ρ' is in the closure of the isomorphism class of ρ if ρ' is a limit of conjugates of ρ . With the notation of 2.9 (respectively 2.14), this means that for each character χ , $\nu_\rho(\chi)$ and $\nu_{\rho'}(\chi)$ have the same dimension $r(\chi)$, and that the unipotent conjugacy class of $\mathrm{GL}(r(\chi))$ defined by the partition $\lambda_{\rho'}(\chi)$ of $r(\chi)$ is in the closure of the one defined by $\lambda_\rho(\chi)$.

Transporting this definition to the moderately ramified $\overline{\mathbb{Q}}_\ell$ -sheaves on $X_{(s)}^*$, define $\overline{E}(\mathcal{R})$ as the set of classes, up to semisimplification, of smooth $\overline{\mathbb{Q}}_\ell$ -sheaves $\mathcal{F}$ of rank r on $X - S$ such that for each $s \in S$, $\mathcal{F}|_{X_{(s)}^*}$ is in the closure of the isomorphism class of $\mathcal{R}(s)$. Define $\overline{M}_{\mathbb{C}}(\mathcal{R}^*)$ as the moduli space of local systems of rank r on $\Sigma - S$, whose local monodromy at s is in the closure of $\mathcal{R}^*(s)$. These local systems are taken up to semisimplification.

We still have $V : \overline{E}(\mathcal{R}) \rightarrow \overline{E}(\mathcal{R})$, and we may repeat conjecture 2.15 for $\overline{E}(\mathcal{R})$ and for the corresponding $\overline{M}_{\mathbb{C}}(\mathcal{R}^*)$.

In some cases, the formulas giving the number of fixed points of V become simpler if $E(\mathcal{R})$ is replaced by $\overline{E}(\mathcal{R})$. See 4.5.

The precisions 2.17 to 2.22 that follow, added in July 2013, owe much to the talks that I gave at the IHES in March and April 2013.

2.17. Let us first place ourselves in the everywhere unramified case: S is empty, and E_r is the set of isomorphism classes of irreducible smooth $\overline{\mathbb{Q}}_\ell$ -sheaves of rank r on X . Let $W(\mathbf{F})$ be the ring of Witt vectors of $\mathbf{F}$, let K be its field of fractions, and let $\overline{K}$ be an algebraic closure of K . Let X_W be a lifting of X over W , and write X_K (respectively $X_{\overline{K}}$) for the curve over K (respectively $\overline{K}$) obtained from it. Let M_K (respectively $M_{\overline{K}}$) be the scheme over K (respectively $\overline{K}$) which is the moduli space of geometrically irreducible vector bundles with connection of rank r on X_K (respectively $X_{\overline{K}}$). By extension of scalars from K to $\overline{K}$, M_K becomes $M_{\overline{K}}$. Let ι be an embedding of $\overline{K}$ in $\mathbb{C}$, and, in 2.14, take for Σ the Riemann surface deduced from $X_{\overline{K}}$ by extension of scalars from $\overline{K}$ to $\mathbb{C}$. For this choice of Σ , the space $M_{\mathbb{C}}$ corresponding to E_r , defined in 2.14, is obtained from $M_{\overline{K}}$ by extension of scalars from $\overline{K}$ to $\mathbb{C}$, and for every prime number ℓ' , the ℓ' -adic cohomology of $M_{\overline{K}}$ is therefore identified with the cohomology of $M_{\mathbb{C}}$ with coefficients in $\mathbb{Q}_{\ell'}$. The conjectures 2.1 (i), (ii) (for S empty) are therefore consequences of the following.

Conjecture 2.18. *The cohomology of $M_{\overline{K}}$ admits an endomorphism “which deserves to be called V^* ” such that for each $n \geq 1$, the number N_n of fixed points of V^n acting on E_r is given by*

$$N_n = \sum_i (-1)^i \mathrm{Tr}(V^{*n}, H^i(M_{\overline{K}})). \quad (2.18.1)$$

In (2.18.1), H^i is ℓ' -adic cohomology for any prime number ℓ' . One should also be able to take the de Rham cohomology of M_K/K . The phrase in quotation marks has no precise meaning, but suggests compatibilities. For example, V^* should be an endomorphism of the cohomology algebra.

There will not in general exist an endomorphism $V : M_{\overline{K}} \rightarrow M_{\overline{K}}$ inducing V^* on cohomology, but I hope that there exists, in the analytic space (in the sense of Berkovich) M_K^{an} associated with $M_{\overline{K}}$, an open subset M_K^0 such that

- (a) the restriction morphism $H^*(M_{\overline{K}}) = H^*(M_{\overline{K}}^{\text{an}}) \rightarrow H^*(M_{\overline{K}}^0)$ is an isomorphism, and
- (b) a crystalline interpretation defines $V = \text{Frob}^* : M_{\overline{K}}^0 \rightarrow M_{\overline{K}}^0$ inducing V^* on cohomology.

The conjectural morphism $V : M_{\overline{K}}^0 \rightarrow M_{\overline{K}}^0$ should send $M_{\overline{K}}^0$ into a proper open subset of $M_{\overline{K}}^0$, corresponding to vector bundles with connection for which parallel transport has a larger radius of convergence; this is what makes natural the choice in (2.18.1) of ordinary cohomology, rather than cohomology with compact support.

A variant of conjecture 2.18, for the completions 2.16 of E_r and $M_{\overline{K}}$, might be more natural.

Example 2.19. Suppose that $r = 1$. In this case, $M_{\overline{K}}$ is the universal additive extension of the Jacobian $\text{Pic}^0(X_{\overline{K}})$, and

$$H^* \text{Pic}^0(X_{\overline{K}}) \xrightarrow{\sim} H^*(M_{\overline{K}}) \quad (2.19.1)$$

is an isomorphism. Moreover one has a natural isomorphism

$$H^*(\text{Pic}^0(X)) \xrightarrow{\sim} H^*(\text{Pic}^0(X_{\overline{K}})). \quad (2.19.2)$$

Over $\mathbf{F}$, one has $V : \text{Pic}^0(X) \rightarrow \text{Pic}^0(X)$, obtained functorially from $\text{Frob} : X \rightarrow X$ (cf. 2.2). This endomorphism V of $\text{Pic}^0(X)$ gives

$$V^* : H^*(\text{Pic}^0(X)) \rightarrow H^*(\text{Pic}^0(X)). \quad (2.19.3)$$

Define the endomorphism

$$V^* : H^*(M_{\overline{K}}) \rightarrow H^*(M_{\overline{K}}) \quad (2.19.4)$$

as transported from (2.19.3) by (2.19.2) and (2.19.1).

Proposition 2.20. *If $r = 1$ and V^* is defined as above, then (2.18.1) is true.*

Proof for $n = 1$ (one reduces to this case by an extension of scalars from $\mathbf{F}_q$ to $\mathbf{F}_{q^n}$). The endomorphisms Frob and V of $\text{Pic}^0(X)$ are transposes of one another under the autoduality of the Jacobian $\text{Pic}^0(X)$ of X . Hence

$$\text{Tr}(V^*, H^i(\text{Pic}^0(X))) = \text{Tr}(F^*, H^i(\text{Pic}^0(X))), \quad (2.20.1)$$

$$\sum_i (-1)^i \text{Tr}(V^*, H^i(\text{Pic}^0(X))) = |\text{Pic}^0(X_0)(\mathbf{F}_q)|. \quad (2.20.2)$$

Class field theory identifies the set of fixed points of V acting on E_1 with the Pontryagin dual of $\text{Pic}^0(X_0)(\mathbf{F}_q)$, with values in the roots of unity of $\overline{\mathbf{Q}}_\ell$ (see 6.1). Thus

$$N_1 = |\text{Pic}^0(X_0)(\mathbf{F}_q)|$$

and, by (2.20.2), this verifies (2.18.1).

Notice that, again by the autoduality of the Jacobian, the subscheme $\text{Pic}^0(X)^V$ of $\text{Pic}^0(X)$ fixed by V is the Cartier dual of the finite abelian group $\text{Pic}^0(X)^{\text{Frob}} = \text{Pic}^0(X_0)(\mathbf{F}_q)$.

2.21. Let us pass to the tamely ramified case. Choose over W a lifting (X_W, S_W) of X and of its étale divisor S . Let (X_K, S_K) and $(X_{\overline{K}}, S_{\overline{K}})$ be deduced from (X_W, S_W) by extension of scalars from W to K and $\overline{K}$. To define in 2.14 a complex analogue $M_{\mathbb{C}}(\mathcal{R}^*)$ of $E(\mathcal{R})$, we had to make the choices (2.14.1), (2.14.2). To define a p -adic analogue, we shall choose

(2.21.1) an isomorphism between the groups of roots of unity of order prime to p in $\overline{\mathbf{Q}}_\ell$ and in K .

There is no need to give an analogue of (2.14.2), because reduction modulo p gives a canonical isomorphism between the groups of roots of unity of order prime to p in K or $\overline{K}$ and in $\mathbf{F}$.

Let $X_{K(s)}$ be the completion of X_K at (the lifting of) s , and let $X_{K(s)}^* := X_{K(s)} - \{s\}$. The datum (2.21.1) gives a natural isomorphism between the group of finite-order characters $\widehat{\mathbf{Z}}(1) \rightarrow \overline{\mathbf{Q}}_\ell^*$ and the prime-to- p part of $\mathbb{Q}/\mathbb{Z}$. It is exactly what is needed to attach to $\mathcal{R}(s)$ an isomorphism class $\mathcal{R}(s)_K^*$ of vector bundles with connection on $X_{K(s)}^*$. Define $M_K(\mathcal{R}^*)$ to be the scheme over K which is the moduli space of geometrically irreducible vector bundles with connection of rank r on $X_K - S_K$, whose completion at $s \in S$ is isomorphic to $\mathcal{R}(s)_K$.

Let ι be an embedding of K in $\mathbb{C}$. If in 2.14 one takes Σ and $S \rightarrow \Sigma$ deduced from (X_K, S_K) by extension of scalars from K to $\mathbb{C}$, if (2.14.1) is deduced from (2.21.1) by ι , and if (2.14.2) is induced by ι , then $M_{\mathbb{C}}(\mathcal{R}^*)$ defined in 2.14 is obtained from $M_K(\mathcal{R}^*)$ by extension of scalars from K to $\mathbb{C}$.

Let $M_{\overline{K}}(\mathcal{R}^*)$ be deduced from $M_K(\mathcal{R}^*)$ by extension of scalars from K to $\overline{K}$. As in 2.18, I conjecture that the cohomology of $M_{\overline{K}}(\mathcal{R}^*)$ admits an endomorphism “which deserves to be called V^* ” such that, for $n \geq 1$, the number of fixed points of V acting on $E(\mathcal{R})$ is given by

$$N_n = \sum_i (-1)^i \operatorname{Tr}(V^*, H^i(M_{\overline{K}}(\mathcal{R}^*))), \quad (2.21.2)$$

and this conjecture implies 2.15 (i) (ii).

2.22. Continue to suppose that the $\mathcal{R}(s)$ are tamely ramified, and suppose that condition (2.10.2) is satisfied.

Let $\det$ denote the functor “maximal exterior power” $\wedge^r$. It induces a map

$$\det : E(\mathcal{R}) \rightarrow E(\det \mathcal{R}). \quad (2.22.1)$$

Hypothesis (2.10.2) ensures that $\det \mathcal{R}$ satisfies (2.10.1), and that $E(\det \mathcal{R})$ is nonempty, hence a torsor under the group E_1 of isomorphism classes of smooth $\overline{\mathbf{Q}}_\ell$ -sheaves of rank one on X . The action is tensor product.

In 2.16, for $r = 1$ and S empty, we defined a moduli space M_K . We shall denote it here by M_{1K} . The functor $\wedge^r$ induces

$$\det : M_K(\mathcal{R}^*) \rightarrow M_K(\det \mathcal{R}^*), \quad (2.22.2)$$

and $M_K(\det \mathcal{R}^*)$ is a principal homogeneous space under M_{1K} .

Extend scalars from K to $\overline{K}$, and consider the Leray spectral sequence for the morphism $\det$:

$$E_2^{pq} = H^p(M_{\overline{K}}(\det \mathcal{R}^*), \det \mathbf{Q}_\ell) \Rightarrow H^{p+1}(M_{\overline{K}}(\mathcal{R}^*), \mathbf{Q}_\ell).$$

I conjecture the existence of an endomorphism V^* of this spectral sequence, respecting its multiplicative structure. The term E_2 is given by

$$H^p(M_K(\det \mathcal{R}^*)) \otimes H^0(M_K(\det \mathcal{R}^*), R^1 \det \mathbf{Q}_\ell) \xrightarrow{\sim} E_2^{pq},$$

and the cohomologies of $M_{\overline{K}}(\det \mathcal{R}^*)$ and $M_{1\overline{K}}$ are canonically isomorphic. The conjectural existence of V^* therefore implies 6.7.

3. Arinkin’s method

3.1. Let s_0 be a closed point of X_0 . One may identify the closed points s of X both with the $\mathbf{F}$ -points of the $\mathbf{F}$ -scheme X , and with the $\mathbf{F}_q$ -points of the $\mathbf{F}_q$ -scheme X_0 . In this way, the points s

over s_0 are identified with the $\mathbf{F}$ -points of $\mathrm{Spec}(k(s_0))$, that is, with the embeddings of $k(s_0)$ into $\mathbf{F}$ extending the embedding of $\mathbf{F}_q$ into $\mathbf{F}$. If such an embedding s is chosen, a character

$$\varepsilon_0 : k(s_0)^* \longrightarrow \overline{\mathbf{Q}}_\ell^*$$

gives, by 2.9, a character

$$\varepsilon := \varepsilon_0 \mathrm{pr}_s : \widehat{\mathbf{Z}}(1) \longrightarrow k(s_0)^* \longrightarrow \overline{\mathbf{Q}}_\ell^*.$$

If $s = \mathrm{Frob}(s')$, the character ε' of $\widehat{\mathbf{Z}}(1)$ defined by ε_0 and s' is ε^q .

3.2. At each $s_0 \in S_0$, choose a multiset $R(s_0)$ of r characters $k(s_0)^* \rightarrow \overline{\mathbf{Q}}_\ell^*$. By 3.1 this gives, for every $s \in S$, a multiset $R(s)$ of r characters of $\widehat{\mathbf{Z}}(1)$. Let ρ_s be the representation of $\widehat{\mathbf{Z}}(1)$ which is the sum of the one-dimensional representations defined by these characters. The corresponding $\overline{\mathbf{Q}}_\ell$ -sheaves $R(s)$ on the $X_{(s)}^*$ (see 2.9) satisfy (2.1.1). Write R for the family $(R(s))_{s \in S}$.

We shall assume that

$$\prod_s \prod_{\varepsilon \in R(s)} \varepsilon = 1. \quad (3.2.1)$$

By (2.10.1), this is a necessary condition for the existence of a smooth $\overline{\mathbf{Q}}_\ell$ -sheaf $\mathcal{F}$ on $X - S$ such that $\mathcal{F}|_{X_{(s)}^*} \simeq R(s)$. If s_0 has degree d , if ε_0 is a character of $k(s_0)^*$, and if, for each s above s_0 , $\varepsilon[s] := \varepsilon_0 \mathrm{pr}_s$ is the corresponding character of $\widehat{\mathbf{Z}}(1)$, the product of these $\varepsilon[s]$ is the character which 3.1 attaches to $\varepsilon_0^{(q^d-1)/(q-1)} = \varepsilon_0 \circ N_{k(s_0)/\mathbf{F}_q}$ and to any one s above s_0 . It is also the character which 3.1 attaches to $\varepsilon_0|_{\mathbf{F}_q^*}$ and to $\mathbf{F}_q \hookrightarrow \mathbf{F}$. Thus the hypothesis (3.2.1) can be rewritten as

$$\prod_{s_0 \in S_0} \prod_{\varepsilon_0 \in R(s_0)} \varepsilon_0|_{\mathbf{F}_q^*} = 1. \quad (3.2.2)$$

3.3. Suppose moreover that the $R(s)$ satisfy the general-position condition (2.10.3). I learned from Arinkin how, under this hypothesis, to construct an algebraic variety Z over $\mathbf{F}_q$ such that, for every $n \geq 1$, the number $N_n(R)$ of fixed points of the permutation V^n of $E(R)$ is equal to the number of $\mathbf{F}_{q^n}$ -points of Z :

$$N_n(R) = |Z(\mathbf{F}_{q^n})|. \quad (3.3.1)$$

Under the hypotheses of 3.3, conjecture 2.15 (i) is therefore true. One should note that Z will be defined only up to a decomposition into locally closed pieces.

3.4. For each $s_0 \in S_0$, let $\varepsilon[s_0](i)$, $0 \leq i \leq a(s_0)$, be the characters of $k(s_0)^*$ occurring in $R(s_0)$, placed in an arbitrary order. Let $n[s_0](i)$ be the multiplicity of $\varepsilon[s_0](i)$ in the multiset $R(s_0)$, and let $n[s_0]$ be the family of integers $(n[s_0](i))_{0 \leq i \leq a(s_0)}$. For $s \in S$ above s_0 , write $\varepsilon[s](i)$ for the character $\varepsilon[s_0](i) \circ \mathrm{pr}_s$ of $\widehat{\mathbf{Z}}(1)$ (3.1), and set $n[s] := n[s_0]$. The multiset $R(s)$ is

$$R(s) := \{\text{the } \varepsilon[s](i), \text{ with multiplicities } n[s](i)\}. \quad (3.4.1)$$

Let E_0 be a vector bundle of rank r on X_0 . A parabolic structure α_0 of type $(n(s_0))_{s_0 \in S_0}$ on E_0 is the datum, for each $s_0 \in S_0$, of a finite filtration $F(s_0)$ of the $k(s_0)$ -vector space E_{s_0} , such that $\mathrm{Gr}_i^{F(s_0)}(E_{s_0})$ has dimension $n[s_0](i)$. Parabolic structures are defined similarly for E on X . If E_0 on X_0 is provided with a parabolic structure α_0 of type $(n(s_0))_{s_0 \in S_0}$, its inverse image E on X inherits a parabolic structure α of type $(n(s))_{s \in S}$. We say that (E_0, α_0) is geometrically indecomposable if its inverse image E on X admits no nontrivial decomposition $E = E' \oplus E''$ compatible with the parabolic structure α , that is, inducing decompositions of the $F^i E_s$. Let $\mathrm{End}(E_0, \alpha_0)$ (respectively $\mathrm{End}(E, \alpha)$) be the algebra of endomorphisms of E_0 (respectively E) respecting the filtrations $F(s_0)$

of the E_{s_0} (respectively $F(s)$ of the E_s). Each of the following conditions is equivalent to geometric indecomposability:

$$\text{The algebra } \text{End}(E, \alpha) = \text{End}(E_0, \alpha_0) \otimes_{\mathbf{F}_q} \mathbf{F} \text{ has no idempotent other than 0 and 1.} \quad (3.4.2)$$

$$\text{The quotient of the algebra } \text{End}(E_0, \alpha_0) \text{ by its radical is reduced to } \mathbf{F}_q. \quad (3.4.3)$$

Fix an integer d . Consider vector bundles E_0 of rank r and degree d on X_0 , equipped with a parabolic structure α of type $(n[s_0])_{s_0 \in S_0}$. Let $Z(\mathbf{F}_q)$ be the set of isomorphism classes of such (E_0, α) which are geometrically indecomposable. Define $Z(\mathbf{F}_{q^n})$ similarly, replacing $X_0/\mathbf{F}_q$ by $X_0 \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}/\mathbf{F}_{q^n}$. I learned the following theorem from Arinkin.

Theorem 3.5. *Under the hypotheses of 3.2 and 3.3, and with the notation of 3.4, the number $N_n(R)$ of fixed points of V^n (2.1) is*

$$N_n(R) = |Z(\mathbf{F}_{q^n})|. \quad (3.5.1)$$

After replacing X_0 by $X_0 \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}$, if necessary, one is reduced to the case $n = 1$.

One should note that the proof does not suggest a natural bijection between $E(R)^V$ and $Z(\mathbf{F}_q)$. It gives a family of one-dimensional vector spaces indexed by $E(R)^V$, with direct sum V , another family, with sum V' , indexed by $Z(\mathbf{F}_q)$, and an isomorphism of vector spaces from V to V' .

3.6. *Sketch of proof of 3.5 $\Rightarrow$ (3.3.1).* For any scheme T over $\mathbf{F}_q$, put $X_T := X_0 \times_{\mathbf{F}_q} T$ and $S_T := S_0 \times_{\mathbf{F}_q} T$. Define $Z(T)$ to be the set of isomorphism classes of vector bundles of rank r on X_T , equipped with a parabolic structure along S_T , such that the conditions of 3.2 and 3.3 are satisfied on each geometric fibre of $X_T \rightarrow T$: degree d , type of the parabolic structure, and indecomposability.

Naively, one would like to take for the scheme Z of (3.3.1) a scheme representing the functor Z . For several reasons this functor is not representable:

- (a) Since the objects under consideration have nontrivial automorphisms, for instance homotheties, this functor is not a sheaf, even for the Zariski topology. For the Picard functor one bypasses the same problem by passing to the associated sheaf. Here the situation is worse, because, for (E_T, α_T) a vector bundle with parabolic structure on X_T , and for (E_t, α_t) the induced bundle on the fibre X_t of $X_T \rightarrow T$ at $t \in T$,
- (b) the dimension of the automorphism group of (E_t, α_t) , equal to $\dim \text{End}(E_t, \alpha_t)$, is not always locally constant in t ;
- (c) the set of t for which (E_t, α_t) is geometrically indecomposable is not always locally closed.

Because of (c), one cannot even hope that Z comes from an algebraic stack C , in the sense that $Z(T)$ would be the set of isomorphism classes in $C(T)$.

However,

- (d) the automorphism groups are connected algebraic groups. If $k' \subset k''$ are finite extensions of $\mathbf{F}_q$, this ensures that

$$Z(k') \xrightarrow{\sim} Z(k'')^{\text{Gal}(k''/k')}.$$

- (e) In (b) above, the function $t \mapsto \dim \text{End}(E_t, \alpha_t)$ is constructible; in (c), the set of t such that (E_t, α_t) is geometrically indecomposable is constructible.

The properties (d) and (e) make it possible, by “cutting Z into pieces” and taking the associated representable sheaves, to obtain the Z promised by (3.3.1).

3.7. *Sketch of proof of 3.5 (for $n = 1$).* We apply the strategy of 2.5. We must show that $|Z(\mathbf{F}_q)|$ is the number of cuspidal automorphic representations π of $\mathrm{GL}(r, \mathbb{A})$, counted modulo $\mathbf{F}_q$ -torsion, which are unramified outside S_0 and whose local component π_{s_0} , for $s_0 \in S_0$, is of the following type. By the local Langlands correspondence, the representation π_{s_0} of $\mathrm{GL}(r, K_{s_0})$ corresponds to a Frobenius-semisimple $\overline{\mathbf{Q}}_\ell$ -representation ρ of the Weil group $W(\overline{K}_{s_0}/K_{s_0})$. We want, first, ρ to be semisimple (in the language of the Weil-Deligne group: $N = 0$; in Arthur’s language: the local $\mathrm{SL}(2)$ or $\mathrm{SU}(2)$ acts trivially). The group $W(\overline{K}_{s_0}/K_{s_0})^{\mathrm{ab}}$ is $K_{s_0}^*$. Its inertia group $\mathcal{O}_{s_0}^*$ has $k(s_0)^*$ as a quotient. We want the restriction of ρ to inertia to factor through the representation of $k(s_0)^*$ which is the sum of the ε_0 in $R(s_0)$.

Let $P[s_0] \subset \mathrm{GL}(r, \mathcal{O}_{s_0})$ be the parahoric subgroup which is the inverse image of the standard parabolic $\overline{P}[s_0]$ of $\mathrm{GL}(r, k(s_0))$ of type $n(s_0)$. The reductive quotient $L[s_0]$ of $\overline{P}[s_0]$ is $\prod_i \mathrm{GL}(n[s_0](i), k(s_0))$. Let $U[s_0]$ be the kernel of $P[s_0] \rightarrow L[s_0]$, and let $\varepsilon[s_0]$ be the character

$$\varepsilon[s_0] = \prod_i \varepsilon[s_0](i)(\det g_i) : P[s_0] \rightarrow \overline{P}[s_0] \rightarrow L[s_0] \rightarrow \overline{\mathbf{Q}}_\ell^*.$$

of the parahoric $P[s_0]$. By construction, the $\varepsilon[s_0](i)$ are pairwise distinct. We shall admit the following assertion.

Assertion 3.8. *The irreducible admissible representations of $\mathrm{GL}(r, K_{s_0})$ of the type defined in 3.7 are those admitting a line stable under P , on which P acts by the character $\varepsilon[s_0]$. This line is unique.*

3.9. Let $\mathcal{A}'$ be the set of cuspidal automorphic representations π of $\mathrm{GL}(r, \mathbb{A})$ admitting a line $L(\pi)$ fixed by the $\mathrm{GL}(r, \mathcal{O}_x)$ for $x \notin S_0$, and stable under $P[s_0]$ for $s_0 \in S_0$, the action of $P[s_0]$ being given by $\varepsilon[s_0]$. The line $L(\pi)$ is unique if it exists. By Lafforgue [L] and 3.8, $N_1(R)$ is the number of classes modulo $\mathbf{F}_q$ -torsion in $\mathcal{A}'$.

Fix an idèle d of valuation 1, and let $\mathcal{A}$ be the set of π in $\mathcal{A}'$ whose central character ω_π takes the value 1 at d . As observed after 2.3.1, one has

$$N_1(R) = |\mathcal{A}' \bmod \mathbf{F}_q\text{-torsion}| = |\mathcal{A} \bmod \mathbf{F}_q\text{-torsion by } \zeta \in \mu_r(\overline{\mathbf{Q}}_\ell)| = \frac{1}{r} |\mathcal{A}|. \quad (3.9.1)$$

3.10. Let A be the vector space of functions on $\mathrm{GL}(r, \mathbb{A})$ which are

- (a) invariant on the left under $\mathrm{GL}(r, K) d^{\mathbb{Z}}$;
- (b) invariant on the right under $\mathrm{GL}(r, \mathcal{O}_x)$, for $x \notin S_0$;
- (c) transformed by $\varepsilon[s_0]$ for the right action of $P[s_0]$, for $s_0 \in S_0$.

A key point in the proof of 3.5 is the following lemma.

Lemma 3.11. *The vector space A is the direct sum of the lines $L(\pi)$, for π in $\mathcal{A}$. It therefore has dimension $rN_1(R)$.*

This lemma is the automorphic counterpart, and a consequence, of the fact that the irreducibility of the $\overline{\mathbf{Q}}_\ell$ -sheaves F_0 under consideration follows from the local hypotheses imposed. One must check that the functions in A are automatically cuspidal. Once this is done, if one writes the space of cuspidal functions on $\mathrm{GL}(r, K) d^{\mathbb{Z}} \backslash \mathrm{GL}(r, \mathbb{A})$ as a direct sum of irreducible representations π of $\mathrm{GL}(r, \mathbb{A})$, the π in $\mathcal{A}$ contribute $L(\pi)$ to A , and the others do not contribute. The cuspidality

of the functions in A follows from the fact that in the spectral decomposition of functions on $\mathrm{GL}(r, K)d^{\mathbb{Z}} \backslash \mathrm{GL}(r, \mathbb{A})$, the induced representations giving rise to Eisenstein series have no nonzero vector satisfying 3.10 (b) (c).

3.12. The space A , the sum of the $L(\pi)$ for $\pi \in \mathcal{A}$, is identified with the space of functions f on

$$\mathrm{GL}(r, K)d^{\mathbb{Z}} \backslash \mathrm{GL}(r, \mathbb{A}) \Big/ \left(\prod_{x \notin S_0} \mathrm{GL}(r, \mathcal{O}_x) \prod_{s_0 \in S_0} U[s_0] \right) \quad (3.12.1)$$

such that, for each $s_0 \in S_0$, the right action of $P[s_0]$ satisfies $f(gp) = \varepsilon[s_0](p)f(g)$.

Let v be the map from (3.12.1) to $\mathbb{Z}/r$ induced by $v \det : \mathrm{GL}(r, \mathbb{A}) \rightarrow \mathbb{Z}$. Let $\overline{\mathcal{A}}$ be the quotient of $\mathcal{A}$ by the action by $\mathbf{F}_q$ -torsion of $\mu_r(\overline{\mathbf{Q}}_\ell)$, and for each orbit B , let V_B be the subspace of A given by the sum of the $L(\pi)$ for π in B . By (2.3.1), $\dim(V_B) = r$. For $m \in \mathbb{Z}/r$, write $(3.12.1)_m$ for the part of (3.12.1) where $v = m$, φ_m for its characteristic function, and $L(\pi, m) = \{\varphi_m f \mid f \in L(\pi)\}$. The dimension of $L(\pi, m)$ is ≤ 1 . For each character η of $\mathbb{Z}/r$, if π' is the $\mathbf{F}_q$ -twist of π by $\eta(1)$, multiplication by η^v transforms $L(\pi)$ into $L(\pi')$. For $\pi \in B$, $L(\pi, m)$ is therefore independent of π ; write $L(B, m) := L(\pi, m)$. The space V_B is stable under multiplication by the η^v , hence under multiplication by the φ_m : for $\pi \in B$,

$$V_B = \bigoplus_{m \in \mathbb{Z}/r} L(B, m).$$

Thus each $L(B, m)$ has dimension one.

The sum A_m of the $L(B, m)$ has dimension $|\overline{\mathcal{A}}| = N_1(R)$. It is identified with the space of functions f on $(3.12.1)_m$ such that, for each $s_0 \in S_0$, the right action of $P[s_0]$ satisfies $f(gp) = \varepsilon[s_0](p)f(g)$. Let $\mathrm{GL}(r, \mathbb{A})^{(d)}$ be the set of $g \in \mathrm{GL}(r, \mathbb{A})$ such that $v \det(g) = d$. If $\bar{d}$ is the class of d modulo r , the set

$$\mathrm{GL}(r, K) \backslash \mathrm{GL}(r, \mathbb{A})^{(d)} \Big/ \left(\prod_{x \notin S_0} \mathrm{GL}(r, \mathcal{O}_x) \prod_{s_0 \in S_0} U[s_0] \right) \quad (3.12.2)$$

maps bijectively to $(3.12.1)_{\bar{d}}$. Therefore:

Lemma 3.13. $N_1(R)$ is the dimension of the space, again denoted A_d , of functions f on (3.12.2) which, for the right action of the $P[s_0]$, $s_0 \in S_0$, satisfy $f(gp) = \varepsilon[s_0](p)f(g)$.

3.14. The set (3.12.2) is interpreted as the set of isomorphism classes of triples (E_0, α, β) , where E_0 is a vector bundle of rank r and degree d on X_0 , and where α and β are structures of the following types on E_0 :

- (a) α is a parabolic structure of type $(n[s_0])_{s_0 \in S_0}$;
- (b) β is a choice of bases of the vector spaces $\mathrm{Gr}_i^{F(s_0)}(E_{s_0})$.

We identify a basis as in (b) with an isomorphism

$$\beta_{s_0, i} : k(s_0)^{n[s_0](i)} \xrightarrow{\sim} \mathrm{Gr}_i^{F(s_0)}(E_{s_0}).$$

With this interpretation, the space A_d of 3.13 becomes the space of functions $f(E_0, \alpha, \beta)$, for E_0, α, β as above, such that

- (i) $f(E_0, \alpha, \beta)$ depends only on the isomorphism class of (E_0, α, β) ;

(ii) for $s_0 \in S_0$ and $g \in L[s_0] = \prod_i \mathrm{GL}(n[s_0](i), k(s_0))$, one has

$$f(E_0, \alpha, \beta g) = \varepsilon[s_0](g) f(E_0, \alpha, \beta).$$

For T an automorphism of (E_0, α) , let $\mathrm{Gr}_i^{F(s_0)}(T_{s_0})$ denote the automorphism of $\mathrm{Gr}_i^{F(s_0)}(E_{0,s_0})$ induced by T , and put

$$\chi(T) := \prod_{s_0} \prod_i \varepsilon[s_0](i) (\det \mathrm{Gr}_i^{F(s_0)}(T_{s_0})). \quad (3.14.1)$$

For every β , hypothesis (i) gives $f(E_0, \alpha, T(\beta)) = f(E_0, \alpha, \beta)$. Hypothesis (ii) gives $f(E_0, \alpha, T(\beta)) = \chi(T) f(E_0, \alpha, \beta)$. If there exists an automorphism T such that $\chi(T) \neq 1$, then $f(E_0, \alpha, \beta) = 0$. Conversely, if $\chi(T) = 1$ for every automorphism T of (E_0, α) , then f may be prescribed freely at one (E_0, α, β) , and the value at (E_0, α, β) determines the values at the (E_0, α, β') .

The dimension of A_d is therefore the number of isomorphism classes of (E_0, α) as above such that $\chi(T) = 1$ for every automorphism T . Theorem 3.5 follows from the characterization (3.4.3) of geometric indecomposability and from the following proposition.

Proposition 3.15. *Let E_0 be a vector bundle of rank r on X_0 , endowed with a parabolic structure of type $(n(s_0))_{s_0 \in S_0}$. The following conditions are equivalent:*

- (i) the quotient of the $\mathbf{F}_q$ -algebra $\mathrm{End}(E_0, \alpha)$ by its radical is reduced to $\mathbf{F}_q$;
- (ii) for every automorphism T of (E_0, α) , $\chi(T)$, defined by (3.14.1), is equal to 1.

We shall deduce that (i) implies (ii) from (3.2.2), and that the negation of (i) implies the negation of (ii) from the general-position hypothesis (2.7.3).

Proof of (i) $\Rightarrow$ (ii). If λ is the image in $\mathbf{F}_q^*$ of the automorphism $T \in \mathrm{End}(E_0, \alpha)^*$, then

$$\varepsilon[s_0](i) (\det \mathrm{Gr}_i^{F(s_0)}(T_{s_0})) = \varepsilon[s_0](i)^{n[s_0](i)}(\lambda),$$

and therefore (3.2.2) implies $\chi(T) = 1$.

Proof of (ii) $\Rightarrow$ (i). The quotient of the $\mathbf{F}_q$ -algebra $\mathrm{End}(E_0, \alpha)$ by its radical is a product of matrix algebras over finite extensions of $\mathbf{F}_q$. If it is not reduced to $\mathbf{F}_q$, it contains a commutative separable $\mathbf{F}_q$ -algebra of dimension > 1 . This algebra has a lifting to $\mathrm{End}(E_0, \alpha)$.

Suppose therefore that $\mathrm{End}(E_0, \alpha)$ contains a commutative separable subalgebra k of dimension > 1 . We must show that the character (3.14.1) of $\mathrm{End}(E_0, \alpha)^*$ is nontrivial. We shall prove that its restriction χ to k^* is nontrivial.

Split case. Suppose first that k is a product of copies of $\mathbf{F}_q$, say $k \simeq \mathbf{F}_q^I$, and that the closed points $s_0 \in S_0$ have degree 1: $\mathbf{F}_q \xrightarrow{\sim} k(s_0)$. The $\mathbf{F}_q^I$ -module structure of E_0 gives a decomposition

$$E_0 = \bigoplus_{\iota \in I} E_0^\iota \quad (3.15.1)$$

such that $(\lambda_\iota)_{\iota \in I} \in \mathbf{F}_q^I$ acts on E_0^ι by multiplication by λ_ι . The decomposition (3.15.1) is compatible with the parabolic structure. For $\iota \in I$, set

$$n_\iota[s_0](i) = \dim \mathrm{Gr}_i^{F(s_0)}((E_0^\iota)_{s_0}).$$

The sum over i of the $n_\iota[s_0](i)$ is the rank of the bundle E_0^ι . The restriction of χ to the factor of index ι of $(\mathbf{F}_q^I)^* = \prod_{\iota \in I} \mathbf{F}_q^*$ is

$$\lambda \mapsto \prod_{s_0 \in S_0} \prod_i \varepsilon[s_0](i)^{n_\iota[s_0](i)}(\lambda). \quad (3.15.2)$$

Its composite with the morphism $\widehat{\mathbf{Z}}(1) \rightarrow \mathbf{F}_q^*$ of 3.1 is the character

$$\prod_{s \in S} \prod_i \varepsilon[s](i)^{n_{\iota[s_0]}(i)}$$

of $\widehat{\mathbf{Z}}(1)$. Hypothesis (2.7.3) ensures that this character is nontrivial.

Reduction of the general case to the split case. It suffices to show that the nontriviality of χ is invariant under extension of scalars from $\mathbf{F}_q$ to a finite extension $\mathbf{F}_{q^n}$. After such an extension of scalars, X_0 becomes a curve X' over $\mathbf{F}_{q^n}$, S_0 becomes a divisor S' of X' , and k becomes $k' := k \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}$. For $s' \in S'$ above $s_0 \in S_0$, put $n[s'] := n[s_0]$, and let $\varepsilon[s'](i)$ be the composite of $\varepsilon[s_0](i)$ with the norm $N_{k(s')/k(s_0)}$. The bundle E' on X' , the inverse image of E_0 , inherits a parabolic structure α' of type $(n(s'))_{s' \in S'}$. Repeating construction (3.14.1) for E' on $X'/\mathbf{F}_{q^n}$, one obtains a character χ' of k'^* . The curve $X/\mathbf{F}$, the integers $n[s](i)$, and the characters $\varepsilon[s](i)$ of $\widehat{\mathbf{Z}}(1)$ all come equally from $X_0/\mathbf{F}_q$ and from $X'/\mathbf{F}_{q^n}$.

To prove that the triviality of χ is equivalent to that of χ' , it is enough to prove that

$$\chi' = \chi \circ N_{k'/k}. \quad (3.15.3)$$

Indeed, the norm $N_{k'/k}$ is surjective. Identity (3.15.3) follows from identity (3.16.1) below, applied to each $\mathrm{Gr}_i^{F(s_0)}(E_{s_0})$, with its structure of $(k, k(s_0))$ -bimodule.

Let k_0 be a finite extension of $\mathbf{F}_q$, let N_0 be a finite-dimensional k_0 -vector space, let k be a product of finite extensions of $\mathbf{F}_q$, and let $\rho : k \rightarrow \mathrm{End}_{k_0}(N_0)$. The (k, k_0) -bimodule N defines a homomorphism

$$[N] : k^* \longrightarrow k_0^* : \lambda \longmapsto \det(\rho(\lambda)).$$

Extend scalars from $\mathbf{F}_q$ to $\mathbf{F}_{q^n}$. We obtain k'_0 , k' , and a (k', k'_0) -bimodule N' . This bimodule defines

$$[N'] : k'^* \longrightarrow k'_0^* : \lambda \longmapsto \det_{k'_0}(\rho(\lambda)).$$

Lemma 3.16. *The diagram*

$$\begin{array}{ccc} k'^* & \xrightarrow{[N']} & k'_0^* \\ N_{k'/k} \downarrow & & \downarrow N_{k'_0/k_0} \\ k^* & \xrightarrow{[N]} & k_0^* \end{array} \quad (3.16.1)$$

is commutative.

Proof. Extend scalars from $\mathbf{F}_q$ to $\mathbf{F}$. Diagram (3.16.1) embeds in the analogous diagram (3.16.2), where k and k_0 are replaced by $k_F := k \otimes_{\mathbf{F}_q} \mathbf{F}$ and $k_{0F} := k_0 \otimes_{\mathbf{F}_q} \mathbf{F}$, where N_0 is replaced by a (k_F, k_{0F}) -bimodule N , and where the extension $\mathbf{F}_{q^n}$ of $\mathbf{F}_q$ is replaced by a product $\mathbf{F}^J$ of n copies of $\mathbf{F}$.

$$\begin{array}{ccc} (k_F \otimes \mathbf{F}^J)^* & \xrightarrow{[N \otimes \mathbf{F}^J]} & (k_{0F} \otimes \mathbf{F}^J)^* \\ \downarrow & & \downarrow \\ k_F^* & \xrightarrow{[N]} & k_{0F}^* \end{array} \quad (3.16.2)$$

One has $(k_F \otimes \mathbf{F}^J)^* = (k_F^*)^J$, $(k_{0F} \otimes \mathbf{F}^J)^* \simeq (k_{0F}^*)^J$, and the vertical “norm” maps are the product maps. The first horizontal row is the product, indexed by J , of copies of the second horizontal row. The commutativity of 3.16.2 follows.

4. The trace formula.

4.1. In [DF], counting problem 2.3 (i) is solved when $|S_0| \geq 2$ and the prescribed local monodromies are unipotent with a single Jordan block. In automorphic language, this means that one requires the local components $\pi_{s_0}, s_0 \in S_0$, to be of the form

$$\text{special representation} \otimes \chi \det g, \quad (4.1.1)$$

where χ is an unramified character of $K_{s_0}^*$.

The hypothesis $|S_0| \geq 2$ allows one to pass from automorphic representations for $\text{GL}(r, K)$ to automorphic representations for the multiplicative group of a division algebra D of dimension r^2 over K . It is reassuring to use the trace formula only in a case with compact quotient, modulo the center, but it should be possible instead to use the trace formula for $\text{GL}(n)$, for a suitable test function, with the simplification brought by the fact that one wants to detect automorphic representations which at two places are in the discrete series. After all, this is how one relates $\text{GL}(r, K)$ and D^* . We shall simplify the explanations which follow by remaining with $\text{GL}(r, K)$, but admitting that the only nonzero terms in the required trace formula are those associated with conjugacy classes of elliptic elements of finite order in $\text{GL}(r, K)$.

Choose an embedding $\mathbf{F}_{q^r} \hookrightarrow M_r(\mathbf{F}_q)$. It defines a morphism

$$\mathbf{F}_{q^r}^* \hookrightarrow \text{GL}(r, \mathbf{F}_q) \hookrightarrow \text{GL}(r, K),$$

by which the set of orbits of $\text{Gal}(\mathbf{F}_{q^r}/\mathbf{F}_q)$ in $\mathbf{F}_{q^r}^*$ maps bijectively onto the set of conjugacy classes to be considered.

Let $T(\gamma)$ be the term of the trace formula, giving the number of fixed points of V , thus associated with γ in $\mathbf{F}_{q^r}^*$. It depends only on the subextension $\mathbf{F}_{q^m}$ of $\mathbf{F}_q$ generated by γ . Put $T(m) := T(\gamma)$. The number c_m of elements of $\mathbf{F}_{q^m}^*$ which generate the extension $\mathbf{F}_{q^m}$ of $\mathbf{F}_q$ is

$$c_m = \sum_{a|m} \mu(a)(q^{m/a} - 1).$$

The number of fixed points of V is

$$\begin{aligned} N_1 &= \sum_{m|r} \frac{c_m}{m} T(m) = \sum_{m|r} \sum_{ab=m} \frac{1}{m} \mu(a)(q^b - 1) T(ab) \\ &= \sum_b \sum_{ab|r} \frac{1}{m} \mu(a)(q^b - 1) T(ab). \end{aligned} \quad (4.1.2)$$

The number of fixed points of V^n , $n \geq 1$, is obtained in the same way after extending scalars from $\mathbf{F}_q$ to $\mathbf{F}_{q^n}$.

Surprise 4.2. *As n varies, each $T(m)$ is in general not, as a function of n , of the form (1.2.1). On the other hand, in (4.1.2), each term of the sum over b is of that form.*

I have no explanation, neither geometric nor automorphic, for 4.2, only a proof. Nor do I have an explanation for the following.

Surprise 4.3. *Under the hypotheses of 4.1, the number N_1 of fixed points of V is divisible by q .*

4.4. If $r = 2$, the trace formula is somewhat less frightening. Drinfeld [Dr] used it to treat the case where S_0 is empty. Flicker, unpublished, treated the case where $|S_0| = 1$ and where the local monodromy at each $s \in S$ is unipotent with a single Jordan block, that is, since $r = 2$, nontrivial. Putting together the information thus obtained, one obtains the following.

Surprise 4.5. *Let E be the set of isomorphism classes of irreducible smooth $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on $X - S$, with unipotent local monodromy, both nontrivial and trivial. The number of fixed points of $V := \text{Frob}^*$ on E depends only on $X_0/\mathbf{F}_q$ and on $|S|$.*

This is the surprise to which 2.16 alludes. If, as in [DF], one requires that the local monodromy at each $s \in S$ be unipotent with a single Jordan block, the number of fixed points depends on X_0 , on $|S|$, and also on the action of Frobenius on S , viewed as a conjugacy class in the symmetric group $S_{|S|}$.

5. Examples

5.1. Take $X_0 = \mathbb{P}^1$, $|S| = 4$, $r = 2$, and let E be the set of isomorphism classes of irreducible smooth $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on $X - S$, with nontrivial unipotent local monodromy at each $s \in S$. In this case, the number of fixed points of the iterates of the permutation $V = \text{Frob}^*$ of E , in the notation of 2.1, is given by

$$N_n = q^n. \quad (5.1.1)$$

If $|S_0| \geq 2$, (5.1.1) is an application of the main result of [DF]. The following curious result ([DF], §7) permits one to deduce from it the case where S_0 is reduced to a closed point of degree 4.

Proposition 5.2. *Every permutation σ of type $(2, 2)$ of S extends to a projectivity of $\mathbb{P}^1$. The action of this projectivity on E is trivial.*

The proof is by reduction to an analogous theorem over $\mathbb{C}$.

5.3. It is natural to complete the set E of 5.1 to $\bar{E}$, the set of isomorphism classes up to semisimplification of smooth $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on $X - S$, with unipotent local monodromy at each $s \in S$. Since $X_0 = \mathbb{P}^1$ and $|S| = 4$, this completion $\bar{E}$ differs from E only by adjoining one point corresponding to the constant $\overline{\mathbf{Q}}_\ell$ -sheaf $\overline{\mathbf{Q}}_\ell^2$.

We again write V for the permutation of $\bar{E}$ induced by Frob^* . The number $\bar{N}_n$ of fixed points of V^n acting on $\bar{E}$ is

$$\bar{N}_n = q^n + 1 \quad (\text{fixed points on } \bar{E}). \quad (5.3.1)$$

5.4. The complex analogue of $\bar{E}$ is the moduli space $\bar{M}$ of complex local systems of rank 2 on $\mathbb{P}^1 - S$, for $|S| = 4$, with unipotent local monodromy. These local systems are taken up to semisimplification. If one chooses a base point $o \in \mathbb{P}^1 - S$, the functor $\mathcal{F} \mapsto \mathcal{F}_o$ is an equivalence from local systems to representations of $\pi_1(\mathbb{P}^1 - S, o)$, and one may take as coordinates on $\bar{M}$ the traces of the elements of π_1 . The space $\bar{M}$ is affine, purely of dimension 2, with a unique singular point corresponding to the constant local system $\mathbb{C}^2$.

The singularity is of type D_4 . The nonzero Betti numbers of $\bar{M}$ are

$$b_0 = 1, \quad b_2 = 1. \quad (5.4.1)$$

5.5. I have not made a complete verification, but the situation seems entirely similar for $X_0 = \mathbb{P}^1$, $|S| = 3$, $r = 3$, with unipotent local monodromy. Again, if one of the local monodromies is not a single Jordan block, the $\mathbf{Q}_\ell$ -sheaf is an iterated extension of constant local systems $\mathbf{Q}_\ell$. The analogue of 5.3 holds for σ a cyclic permutation of the three points of S . The number of fixed points of V^n is again $q^n + 1$, with the “1” given by the reducible local systems with unipotent local monodromy, all having a constant local system as semisimplification. The unique singular point of the complex analogue is of type E_6 .

5.6. Take $X_0 = \mathbb{P}^1$, $|S| = 4$, $r = 2$, and suppose that, as in 3.4, the prescribed local monodromy is given at each $s_0 \in S_0$ by two distinct characters $\alpha'[s_0]$, $\alpha''[s_0]$ of $k(s_0)^*$. Suppose (3.2.2) and the general-position hypothesis (2.7.3) are verified. With the aid of (3.5.1), one obtains the following.

Proposition 5.7. *Under the hypotheses of 5.3, one has*

$$N_1 = q + 1 + |S_0(\mathbf{F}_q)|. \quad (5.7.1)$$

Arinkin verified that for $X_0 = \mathbb{P}^1$ and $r = 2$, if a vector bundle E_0 of rank 2 on X_0 with parabolic structure α_0 at S_0 is indecomposable, then $\text{End}(E_0, \alpha)$ is reduced to the scalars. If Z is defined as in 3.6 and one passes to the sheaf associated for the Zariski topology, one obtains a functor representable by an algebraic space. In the case $|S| = 4$, if one takes degree $d = 1$, it is represented by the sum of two copies of X_0 , glued along $X_0 - S_0$. The number of points of this scheme over $\mathbf{F}_q$ is given by (5.7.1).

5.8. Let us pass to the complex analogue. On $\mathbb{P}^1 - S$, with $|S| = 4$, consider the moduli space of complex local systems of rank 2 with prescribed local monodromy at each $s \in S$. At $s \in S$, require that the local monodromy be conjugate to

$$A_s = \begin{pmatrix} a_s & 0 \\ 0 & b_s \end{pmatrix}.$$

Assume $\prod_s a_s b_s = 1$, and that if, for each s , c_s is one of a_s or b_s , one never has $\prod c_s = 1$. Put $A := (A_s)_{s \in S}$, and denote by $M(A)$ the moduli space, 2.10. It is smooth, purely of dimension 2. One may regard $M(A)$, for A_s close to 1, as a deformation of $\bar{M}$ considered in 5.4. In this deformation, the space of vanishing cycles has dimension 4, since the singularity of $\bar{M}$ is of type D_4 . At infinity, the geometry does not change, and hence the nonzero Betti numbers are

$$b_0 = 1, \quad b_2 = 5. \quad (5.4.1)$$

5.9. Take $X_0 = \mathbb{P}^1$, $|S| = 3$, $r = 3$, and suppose that, as in 3.4, the prescribed local monodromy is given at each $s_0 \in S_0$ by three distinct characters of $k(s_0)^*$. Suppose (3.2.2) and the general-position hypothesis 2.7.3 are verified. With the aid of (3.5.1), one obtains here

$$N_1 = q + 1 + 2|S_0(\mathbf{F}_q)|. \quad (5.9.1)$$

The sheaf associated to Z as in 3.6 is indeed, for $d = 1$, represented by the sum of three copies of X_0 , glued along $X_0 - S_0$. If S_0 consists of three rational points over $\mathbf{F}_q$, (5.9.1) says that $N_1 = (q + 1) + 6$, as suggested by the singularity E_6 of the complex analogue for unipotent local monodromy, cf. 5.5.

6. Rank 1

6.1. Let $E(1, \emptyset)$ denote the set $E(\mathcal{R})$ of 2.1 for $r = 1$ and $S_0 = \emptyset$. The tensor product makes it into an abelian group. Class field theory identifies $E(1, \emptyset)^V$ with the group of characters with values in $\bar{\mathbf{Q}}_\ell^*$ of the finite group $\text{Pic}^0(X_0)(\mathbf{F}_q)$. Indeed, this theory identifies the abelianized fundamental group of X_0 with the profinite completion of $\text{Pic}(X_0)$. The group $\text{Pic}(X_0)$ maps to $\mathbb{Z}$ by the degree map, with kernel the finite group $\text{Pic}^0(X_0)(\mathbf{F}_q)$ of $\mathbf{F}_q$ -points of the Jacobian $\text{Pic}^0(X_0)$. The isomorphism classes of smooth $\bar{\mathbf{Q}}_\ell$ -sheaves $\mathcal{F}_0$ of rank one on X_0 are thereby identified with the characters χ of $\text{Pic}(X_0)$ with values in $\bar{\mathbf{Z}}_\ell^*$; $\mathbf{F}_q$ -twisting by λ corresponds to multiplication by the character λ^n of $\mathbb{Z}$, and, assigning to $\mathcal{F}_0$ the restriction of χ to $\text{Pic}^0(X_0)(\mathbf{F}_q)$, one obtains a bijection between $\mathbf{F}_q$ -twist classes of smooth $\mathcal{F}_0$ of rank 1 on X_0 and characters of $\text{Pic}^0(X_0)(\mathbf{F}_q)$.

6.2. For any $r \geq 1$ and $\mathcal{R}$ as in 2.1, tensor product induces an action of the abelian group $E(1, \emptyset)$ on $E(\mathcal{R})$, and of $E(1, \emptyset)^V$ on $E(\mathcal{R})^V$.

If $r = 1$, and $\mathcal{R}$ satisfies the compatibility (2.10.1), class field theory shows that $E(\mathcal{R})^V$ is a principal homogeneous space under $E(1, \emptyset)^V$. Hence

$$|E(\mathcal{R})^V| = |E(1, \emptyset)^V| = |\text{Pic}^0(X_0)(\mathbf{F}_q)| \quad (\text{if } r = 1 \text{ and (2.10.1)}). \quad (6.2.1)$$

If $r > 1$, taking the exterior power $\bigwedge^r$ gives a map

$$\det : E(\mathcal{R}) \longrightarrow E\left(\bigwedge^r \mathcal{R}\right)$$

such that, for the action of $E(1, \emptyset)$, denoted $\otimes$, one has

$$\det(l \otimes \mathcal{F}) = l^r \otimes \det(\mathcal{F}).$$

Similarly after passage to the fixed points of $V = \text{Frob}^*$, for

$$\det : E(\mathcal{R})^V \longrightarrow E\left(\bigwedge^r \mathcal{R}\right)^V$$

and the action of $E(1, \emptyset)^V$.

The action of $E(1, \emptyset)^V$ on $E(\mathcal{R})^V$ is not necessarily free, and the fibers of

$$\det : E(\mathcal{R})^V \longrightarrow E\left(\bigwedge^r \mathcal{R}\right)^V$$

do not necessarily all have the same number of elements. Nevertheless, in all cases which have been computed, $|E(1, \emptyset)^V|$ divides $|E(\mathcal{R})^V|$. As a function of n , $|\text{Pic}^0(X_0)(\mathbf{F}_{q^n})|$ has the form (2.15.1):

$$\begin{aligned} |\text{Pic}^0(X_0)(\mathbf{F}_{q^n})| &= \sum_i (-1)^i \text{Tr}(\text{Frob}^*, H^i(\text{Pic}^0(X), \mathbf{Q}_\ell)) \\ &= \sum_i (-1)^i \text{Tr}(\text{Frob}^*, \bigwedge^i H^1(X, \mathbf{Q}_\ell)). \end{aligned}$$

I conjecture the following more precise form of conjecture 2.15 (i).

Conjecture 6.3. *As a function of n , the number of fixed points $N_n(\mathcal{R})$ of V^n is of the form*

$$N_n(\mathcal{R}) = |\text{Pic}^0(X_0)(\mathbf{F}_{q^n})| \cdot \sum_k c_k \gamma_k^n \quad (6.3.1)$$

for suitable integers c_k and suitable numbers γ_k .

6.4. Here is an analogue of 6.3 in algebraic geometry. Let A_0 be an abelian variety over $\mathbf{F}_q$, and let $f_0 : X_0 \rightarrow A_0$ be a proper smooth morphism. Although the fibers of $f : X_0(\mathbf{F}_{q^n}) \rightarrow A_0(\mathbf{F}_{q^n})$ need not have the same number of elements, for example $X_0 = A_0$ and f_0 multiplication by an integer prime to p , $|A_0(\mathbf{F}_{q^n})|$ divides $|X_0(\mathbf{F}_{q^n})|$. Indeed, the $R^i f_{0!} \mathbf{Q}_\ell$ are smooth $\mathbf{Q}_\ell$ -sheaves on A , and the fundamental group, abelian, of A acts on these sheaves. This action is known to be semisimple. Let $(R^i f_{0!} \mathbf{Q}_\ell)_0$ be the subsheaf of invariants. It comes from a subsheaf $(R^i f_{0!} \mathbf{Q}_\ell)_0$ of $R^i f_{0!} \mathbf{Q}_\ell$ on A_0 , and the latter is the inverse image on A_0 of a sheaf on $\text{Spec}(\mathbf{F}_q)$, which we denote by $H^i(X_0/A_0)$. One knows that

$$H^i(A, (R^j f_{0!} \mathbf{Q}_\ell)_0) \xrightarrow{\sim} H^i(A, R^j f_{0!} \mathbf{Q}_\ell). \quad (6.4.1)$$

Grothendieck's trace formula for the number of $\mathbf{F}_{q^n}$ -points of X_0 can therefore be rewritten as

$$|X_0(\mathbf{F}_{q^n})| = \text{Tr} \left(\text{Frob}^*, \left(\sum_i (-1)^i H^i(A, \mathbf{Q}_\ell) \right) \otimes \left(\sum_j (-1)^j H^j(X/A) \right) \right). \quad (6.4.2)$$

If the γ_k are the eigenvalues of Frobenius on the $H^j(X/A)$, and the c_k are the alternating sum of their multiplicities, one deduces from (6.4.2) that

$$|X_0(\mathbf{F}_{q^n})| = |A_0(\mathbf{F}_{q^n})| \cdot \sum_k c_k \gamma_k^n, \quad (6.4.3)$$

a formula analogous to (6.3.1). The promised divisibility of $|X_0(\mathbf{F}_{q^n})|$ by $|A_0(\mathbf{F}_{q^n})|$ follows from the fact that the eigenvalues of Frobenius on $H^j(X/A)$, contained in the cohomology of a fiber of $X_0 \rightarrow A_0$, are algebraic integers.

6.5. In 6.4, the crucial point is not that f_0 be proper and smooth, but that the $R^i f_{!} \mathbf{Q}_\ell$ be smooth. One may avoid having to suppose that the $R^i f_{!} \mathbf{Q}_\ell$ are semisimple by considering, rather than the subsheaf of invariants for the action of $\pi_1(A)$, the largest subsheaf on which the action is unipotent.

Suppose that there is an action of A_0 on X_0 , and an integer r , such that f_0 is equivariant for the action of A_0 on A_0 given by $x : y \mapsto rx + y$. Under this hypothesis, the inverse images of the $R^i f_{!} \mathbf{Q}_\ell$ on A by multiplication by $r : A \rightarrow A$ are constant sheaves. This implies both their smoothness and their semisimplicity.

6.6. If $g \geq 1$, conjecture 6.3 implies that in the conjectural expression (2.11.1) for $N_n(\mathcal{R})$, one has $\sum a_i = 0$.

Let us pass to the complex analogues, as in 2.10. Let $M(1, \emptyset)$ be the group of isomorphism classes of complex local systems of rank 1 on P . It is isomorphic to $(\mathbb{C}^*)^{2g}$, and, if the compatibility $\prod \det \mathcal{R}^*(s) = 1$ is satisfied, $M(\det \mathcal{R}^*)$ is a principal homogeneous space under $M(1, \emptyset)$. The group $M(1, \emptyset)$ acts on $M(\mathcal{R}^*)$ and on $M(\det \mathcal{R}^*)$. Denote the action by $\otimes$. The map induced by $\bigwedge^r$,

$$\det : M(\mathcal{R}^*) \longrightarrow M(\det \mathcal{R}^*),$$

satisfies $\det(l \otimes \mathcal{F}) = l^r \otimes \det(\mathcal{F})$. This implies that $\det$ is a fibration, that the $R^i \det_! \mathbb{Z}$ are locally constant, and that the action of $\pi_1 M(\det \mathcal{R}^*)$ on the $R^i \det_! \mathbb{Z}$ factors through its quotient

$$\pi_1 / r\pi_1 \simeq (\mathbb{Z}/r\mathbb{Z})^{2g}.$$

If $g \geq 1$, one deduces, by arguments analogous to those of 6.4, that $\chi(M(\mathcal{R}^*)) = 0$, in agreement with 2.11 (ii). Let $(R^i \det_! \mathbb{Q})_0$ be the local subsystem of $R^i \det_! \mathbb{Q}$ of invariants under the action of π_1 , and let $\chi(M(\mathcal{R}^*)/M(\det \mathcal{R}^*))$ be the alternating sum of the ranks of the $(R^i \det_! \mathbb{Q})_0$.

Conjecture 6.7. *With the notation of 6.3 and 6.6, one has*

$$\sum_k c_k = \chi(M(\mathcal{R}^*)/M(\det \mathcal{R}^*)).$$

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